

Calibrated Uncertainty for Active Room-Layout Estimation

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Abstract—A robot that estimates the layout of a room usually reports geometry and little else. Downstream consumers—planners, scene graphs, other agents—must then decide how far to trust it, with nothing to go on. We present an online room-layout estimator whose reported uncertainty is calibrated against ground truth, and which uses that uncertainty as a decision variable: it gates which corners are published, and it is what the robot acts to reduce. Parameters, structure (how many walls there are) and viewpoint selection are all obtained by minimising one variational free energy, so the same quantity in nats prices the data fit, the model complexity and the next observation. On 594 annotated floor plans the estimator reaches 0.983 mean IoU, and—the claim of the paper—a corner it declares publishable has a median true error of 2.0 cm and lies within its declared 6 cm in 86 % of cases, against 3.6 cm and 64 % for corners it withholds. Holding the system to that standard proved diagnostic: it exposed three distinct precision defects in three different modules, all instances of uncertainty that averages away past an error that does not, one of them in a module the instrument was not measuring. We show that no scalar precision can be correct for this problem—the irreducible error varies by $5\times$ between environments and $2\times$ between moving and standing still—and argue that this is why precision must be inferred rather than tuned.

I. INTRODUCTION

Room-layout estimation is mature enough that accuracy alone no longer separates methods. What separates them for a robot is whether the estimate can be *acted on*: whether a planner can tell a wall that is known to a centimetre from one that is a guess, whether a scene graph should commit a corner to shared memory, whether the robot should look again before deciding. All of these are questions about uncertainty, and almost no layout system answers them with a number that has been checked.

This is not an oversight of presentation. An uncertainty that is never validated tends to be wrong, and wrong in a way that is invisible: an estimator that overstates its error is merely timid, while one that understates it is confidently incorrect, and neither shows up in a mean IoU. We found both failure modes in our own system, and we found them only because something downstream was made to depend on the number being right.

Our estimator maintains a room as a cyclic polygon of wall landmarks in Hesse normal form, inferred jointly with the robot pose over a sliding window. Three decisions that are usually taken by three separate mechanisms are here taken by one. *What the walls are* is variational inference over their parameters. *How many walls there are* is Bayesian model comparison: a structural change is adopted when it pays for

its own description length, priced in nats on the same scale as the data fit, rather than by a merge threshold. *Where to look next* is the epistemic term of the expected free energy, so the robot moves to reduce the uncertainty that is currently blocking it from committing to a layout. Because all three are denominated in the same currency, every precision in the model is load-bearing: get one wrong and the comparisons it feeds are priced in a corrupt unit. That is the property we exploit.

Contributions.

- 1) An online layout estimator whose per-corner and per-edge uncertainty is *validated against ground truth* on 594 annotated floor plans, and is *actionable*: it gates publication and drives exploration (§V).
- 2) A propagation from returns to corner covariance that separates the error that averages down with more observation from the error that does not, and a measurement of both terms on real sensor data (§IV).
- 3) Evidence that a scalar precision cannot be calibrated for this problem: the irreducible component varies by $5\times$ across environments and $2\times$ between motion states, which we argue is a general property of map-relative estimation rather than a quirk of our system (§V-D).
- 4) Three precision defects found by calibration, including one localised in a module the statistic was not measuring—offered as evidence that a free-energy formulation earns its keep diagnostically, not just descriptively.

II. RELATED WORK

Floor-plan reconstruction from complete scans. A large family of methods recovers a floor plan from an accumulated point cloud or density image, from the early unified formulation [1] and its shortest-path successor [2] through attention and query-based architectures [3], [4], room-wise implicit and slicing representations [5], [6], room-aware transformers [7], continuity-aware edges [8], guided set diffusion [9], autoregressive polygon sequences [10], and segmentation-guided fusion [11]. They are evaluated by corner and room F1 on a finished scan. Two properties distinguish them from our setting: they are offline, so no decision is taken while the estimate is still uncertain; and the number of corners is chosen by a search with hand-set costs rather than by a likelihood, so there is no quantity in which structure and data fit are commensurate.

Panoramic room layout. Single- and multi-view layout estimation recovers room shape from 360° imagery [12], [13], [14], with self-training and multi-view consistency removing the need for per-scene labels [15], [16], and related work registering panoramas or verifying their alignment [17], [18]. These produce a layout per view rather than a belief a robot accumulates, and report no per-corner uncertainty a consumer could gate on. Unbounded and general (non-Manhattan) variants relax the shape prior [19], [20], [21]. *Sequential* floor-plan estimation from a moving camera [22] is the closest in spirit to ours: it, too, must decide room membership and shape while the scene is still incomplete. Autoregressive scene description [23] likewise produces structure from a walkthrough, as language rather than as a belief with a covariance.

Exploration that predicts structure. A complementary line predicts unobserved layout to drive exploration—room shape from partial observation [24], walls beyond the frontier learned from floor plans [25], information gain from global map predictions [26], and generative layout prediction for planning [27]—together with the question of when a map is finished [28]. These share our premise that a layout belief should inform where to go next; the predicted structure is used to *score* viewpoints rather than to be published with an uncertainty a downstream consumer must trust.

Active SLAM. Surveys of the field [29], [30] formalise viewpoint selection by information-theoretic utility, and the question of whether the usual uncertainty criteria are the right ones has itself been raised [31]. Our selection rule is the epistemic term of an expected free energy, which coincides with information gain under a Gaussian model; the contribution here is not the rule but that the precision it consumes has been verified, so the gain it predicts is a gain that exists.

Uncertainty and calibration. Uncertainty estimation is well studied in dense depth prediction, where predicted confidence can be compared against realised error [32], and in photogrammetric modelling [33], [34]. To our knowledge no online room-layout estimator reports a per-corner covariance that has been validated against layout ground truth, which is the gap this paper addresses. Our position is deliberately parametric: we keep a covariance rather than a coverage set, because a joint estimator and a downstream consumer both need one, and we check it.

III. GENERATIVE MODEL

A room is a cyclic sequence of wall landmarks. Wall w is a line in Hesse normal form, $\mathbf{n}(\phi_w) \cdot \mathbf{p} = d_w$, with belief $\mathcal{N}([\phi_w, d_w]^\top, \Lambda_w^{-1})$. The robot pose over a sliding window of K frames is $x_{1:K}$. Given returns z , perception minimises the variational free energy

$$F = \underbrace{-\log p(z \mid x, \Phi)}_{\text{accuracy}} + \underbrace{D_{\text{KL}}(q(x, \Phi) \parallel p(x, \Phi))}_{\text{complexity}}, \quad (1)$$

which under a Gaussian q reduces to the sparse nonlinear least-squares problem we solve by Gauss–Newton over the window.

Structure as model comparison. The number of walls is not fixed. A candidate structural change—birth, death, or the

splice of a new edge into the cycle—is a different model m' , and is adopted when $F(m') < F(m)$. The complexity term charges each additional edge a fixed description length: with a room of radius R discretised at resolution c , an edge costs $\ln 4 + \ln(2R/c)$ nats to name. There is no merge threshold and no minimum wall length; a thin partition survives exactly when the returns that support it pay for the nats it costs.

Action. A viewpoint v is scored by the expected information gain $\mathbb{E}_o[H(x, \Phi) - H(x, \Phi \mid o)]$, predicted through the model’s own forward sensor: beams are cast against the current belief, and each wall’s expected Fisher contribution $J_w(v)$ gives a Gaussian gain $\frac{1}{2} \ln(\det(\Lambda_w + J_w)/\det \Lambda_w)$. Because publication is gated on corner σ (§IV), the robot acts, in effect, to make its own map publishable.

IV. WHAT A LAYOUT ESTIMATE IS WORTH

A corner is the intersection of two walls, so its covariance follows from theirs. With $M = [\mathbf{n}_a^\top; \mathbf{n}_b^\top]$ and the intersection $\mathbf{p}$,

$$\Sigma_{\mathbf{p}} = J_a \Lambda_a^{-1} J_a^\top + J_b \Lambda_b^{-1} J_b^\top + \sigma_m^2 I, \quad (2)$$

where J ’s ϕ column carries the lever arm $\mathbf{t} \cdot \mathbf{p}$ and M^{-1} contributes $1/\sin \theta$ between the walls, so a shallow corner is correctly reported as uncertain along its bisector without any angle test.

The term that matters for this paper is the last one. The first two shrink as $1/N$ with the returns behind each wall: observe longer and they go to zero. **The true error does not follow them down.** What remains when the statistics are exhausted is systematic—the offset between the mapped wall and the real surface, identical on every look from a given viewpoint—and it is what σ_m represents. Omitting it produces an estimator that grows arbitrarily confident about a corner whose error is bounded below; double-counting it produces one that is timid. We measure it rather than tune it, and §V-D shows why it cannot be a constant.

V. EXPERIMENTS

A. Geometric generality: 594 annotated floor plans

We evaluate on 594 room annotations from **MatterportLayout** [35], the Manhattan room-layout annotations on Matterport3D [36] panoramas, spanning 6 to 18 ground-truth corners.

We state plainly what this does and does not measure. Our observations are *simulated from the annotated polygon*: a range sensor is driven through the annotated room and the estimator never sees a photograph. **The IoU in Table I is therefore not comparable with single-image layout methods evaluated on the same corpus**, which solve a harder problem—recovering shape from appearance—and should not be scored against these numbers in either direction. What the 594 rooms do provide is the *shape distribution of real buildings*: real aspect ratios, real sizes, and non-convex plans with up to 18 corners, in place of a generator’s idea of a room. The quantity under test here is the estimator’s geometry and, in §V-B, its uncertainty— for which the annotated corners are ground truth regardless of how the returns were produced. §V-C then validates the sensor model itself on real depth.

TABLE I
LAYOUT ACCURACY, 594 ANNOTATED FLOOR PLANS, ONE RUN.

IoU, mean	0.9827
IoU, median	0.9870
IoU, 10th percentile	0.9690
IoU ≥ 0.95	95.2 %
IoU ≥ 0.90	98.8 %
Hausdorff, median (m)	0.0705
Coverage of truth interior, mean	0.9996
Failures (no closed polygon)	8 /594

TABLE II
CALIBRATION OF THE PUBLISHED CORNER σ AGAINST GROUND TRUTH.
4344 MATCHED CORNERS. σ IS THE LARGEST-AXIS STANDARD
DEVIATION, SO THIS TESTS A BOUND.

	$\leq 1\sigma$	$\leq 2\sigma$	$\leq 3\sigma$	publishable	claim holds
no σ_m	46.7%	70.5%	81.2%	89.0%	85.8%
$\sigma_m = 0.02$ m	65.3%	85.0%	91.2%	89.0%	86.1%

B. Is the reported uncertainty honest?

For every room we take the final published polygon and match its corners one-to-one to the truth corners by Hungarian assignment. Surplus corners on either side remain unmatched and are counted as *structure* errors—never as position error. This matters: nearest-vertex matching instead yields a mean error of 0.43 m against a median of 0.023 m, because a spurious corner has no truth counterpart and its “error” is meaningless.

Table II shows the effect of the systematic term of (2). Without it, the true error is flat at about 2 cm across every band of declared σ : a corner claiming 1 cm is no more accurate than one claiming 5 cm, and the declared uncertainty ranks corners without predicting their error. The measured floor is 0.019 m, against a simulated sensor of 0.020 m per return. Adding it as a variance term—not a clamp—calibrates the bound at almost no cost in yield.

The actionable claim. Publication is gated at $\sigma \leq 6$ cm. A corner the system declares publishable has a median true error of 2.0 cm and lies within the 6 cm it claims in 86.1 % of cases; a corner it withholds has a median error of 3.6 cm and satisfies the same bound only 63.7 % of the time. The gate discriminates, and the number it gates on now means something.

C. The real-sensor split

The decomposition in (2) claims that error separates into a part that averages away and a part that does not. We measure both on **MP3D-FPE** [22], which provides real Matterport depth with camera trajectories and per-room annotated corners. Slicing the depth to a horizontal band and fitting each annotated wall from the returns that land on it gives both terms directly (588 segments, 120 rooms; the slice registers to the annotations with a median modal offset of 7.5 mm).

Two consequences. First, the simulated sensor used in §V is honest, indeed slightly conservative: real depth scatters at 1.5 cm against the 2.0 cm simulated, so the calibration above

TABLE III
THE TWO COMPONENTS, MEASURED ON REAL DEPTH AGAINST
ANNOTATED WALLS.

	median (m)	p90
scatter about the fitted wall (averages down)	0.0149	0.0382
fitted wall vs. annotated wall (does not)	0.0181	0.0572

rests on a noise model that does not flatter it. Second, the systematic component is a property of *the map*, not of the sensor—which sets up the next result.

D. Why no scalar precision can be correct

On the robot we log, for every corner candidate at the association gate, the innovation and each term of its covariance: 126 008 candidates over a 200 m tour. Normalised innovation squared (NIS, divided by its two degrees of freedom) equals 1 when a covariance matches the error it claims.

Recovering inter-frame motion from the predicted corners themselves (a rigid fit between consecutive frames, so no pose log is needed) reveals that NIS depends on what the robot is *doing* rather than on where it is (Fig. 2). **While the robot moves, the channel is calibrated:** 0.90, 0.97 and 0.92 across three independent rotation bands. It departs from 1 only at the extremes—1.65 at rest and 1.88 under fast rotation—and the detection covariance is flat throughout ($\sigma_{\text{det}} = 0.043$ m), so neither departure belongs to the corner channel.

The two extremes have different causes, and the stationary one is the instructive one. Decomposing the innovation for a fixed corner across stationary frames separates it cleanly: the part that *repeats* is 0.0952 m and the part that *scatters* is 0.0113 m, a ratio of 8.1, with the bias exceeding the scatter in 97 % of cases. Averaging 1.1 cm of genuine scatter earns exactly the 1.56 cm pose uncertainty the estimator reports—it is not mishandling the noise. The remaining 9.5 cm is a map offset, identical on every look from a fixed viewpoint, and represented nowhere in the model: standing still, the estimator accumulates confidence from observations that carry no new information about it.

Combined with Table III, the systematic term measures 1.8 cm against a carefully annotated building and 9.5 cm against a hand-traced layout—a factor of 5 for the same sensor and the same estimator—and varies by a factor of 2 between standing still and moving, because in motion the per-wall offsets partially cancel in the pose solution. A constant tuned in one environment is wrong in the next; a constant tuned while the robot moves is wrong when it stops. This is the argument for representing the offset as an inferred state with a prior rather than a compiled-in number.

E. Calibration as a diagnostic

Requiring precisions to mean what they say found three defects of one kind—uncertainty that averages down past an error that does not. In the layout channel the systematic term was absent (Table II). In the localisation channel it was present twice under two names, making the channel $2.1\times$ too timid;

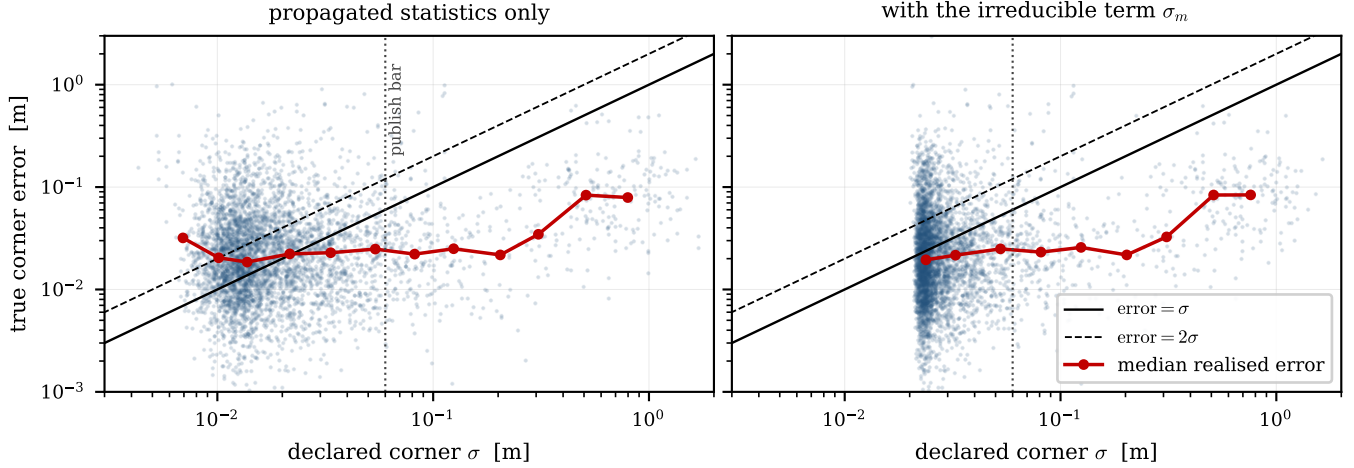


Fig. 1. **Is the declared uncertainty honest?** Every matched corner of 594 annotated floor plans: declared σ against the error it actually made. *Left*, propagating only the walls’ statistical uncertainty: the red median is flat, so a corner claiming 1 cm is no more accurate than one claiming 5 cm, and the cloud sits far above $\text{error} = \sigma$ at the tight end. *Right*, with the irreducible term of (2): the median tracks the diagonal and the bound holds. The dotted line is the publication gate.

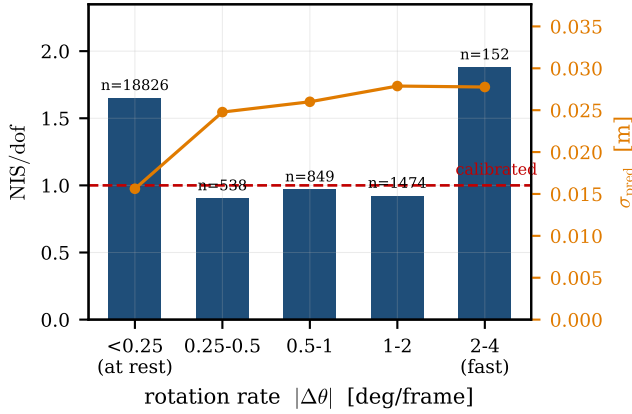


Fig. 2. **The channel is calibrated only while the robot moves.** NIS per rotation band over a 200 m tour, with the pose-induced corner uncertainty σ_{pred} overlaid. Three independent bands of genuine motion sit at 0.90 to 0.97. At rest σ_{pred} collapses while the innovation grows, and under fast rotation the innovation grows again—two distinct mechanisms bracketing a correctly calibrated middle.

removing the duplicate and measuring the survivor moved NIS from 0.47 to 1.03. In the pose estimator it is absent again, and was found by an instrument pointed at the layout. We regard the third as the strongest evidence that the formulation does work rather than description: nothing in a pipeline of separately-tuned heuristics depends on a precision being right, so nothing in it would have flagged any of the three.

VI. LIMITATIONS

Observations in §V are simulated from the annotated polygons; §V-C validates the noise model but not the full pipeline on recorded data. We attempted a full pipeline evaluation on MP3D-FPE [22] and report a negative result we believe

is useful to others. Because it annotates *rooms inside open-plan buildings* while a 360° depth sensor sees through every doorway, a median 57 % of returns fall outside the room being scored (0 of its 401 rooms fall below 10 %); taking the union of a floor’s rooms still leaves 50 % outside, because corridors and stairwells are unannotated. No granularity makes the annotation and the observation describe the same region, so this corpus—and, we expect, any room-annotated building corpus—cannot grade a method that models a *bounded* room. We report it rather than quoting the 0.14 IoU it yields, which measures the task mismatch and not the estimator. Finally, the pose-covariance defect of §V-D is measured but not fixed; the cure we argue for—a per-wall offset state with the measured prior—is future work.

VII. CONCLUSION

A layout estimate that a robot can act on needs an uncertainty a robot can check. We have shown that such an uncertainty can be maintained online, that it can be validated against ground truth at scale, and that it is useful precisely because something depends on it. The discipline of making precisions honest is not bookkeeping: it exposed three defects, and it showed that the quantity everyone tunes as a constant is not one.

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